



TÍTULO DO DOCUMENTO

Subtítulo (introduzir, se aplicável, ou eliminar o campo)

Estudante(s)	Clique para introduzir o(s) nome(s) e número(s) de estudante(s)
Curso	Clique para introduzir o curso
Unidade curricular	Clique para introduzir a designação da unidade curricular
Ano letivo	Clique para introduzir o ano letivo
Docente(s)	Clique para introduzir o(s) nome(s) do(s) docente(s) da unidade curricular
Data	Clique para selecionar a data

DBSCAN ALGORITHM

Dataset1 DBSCAN

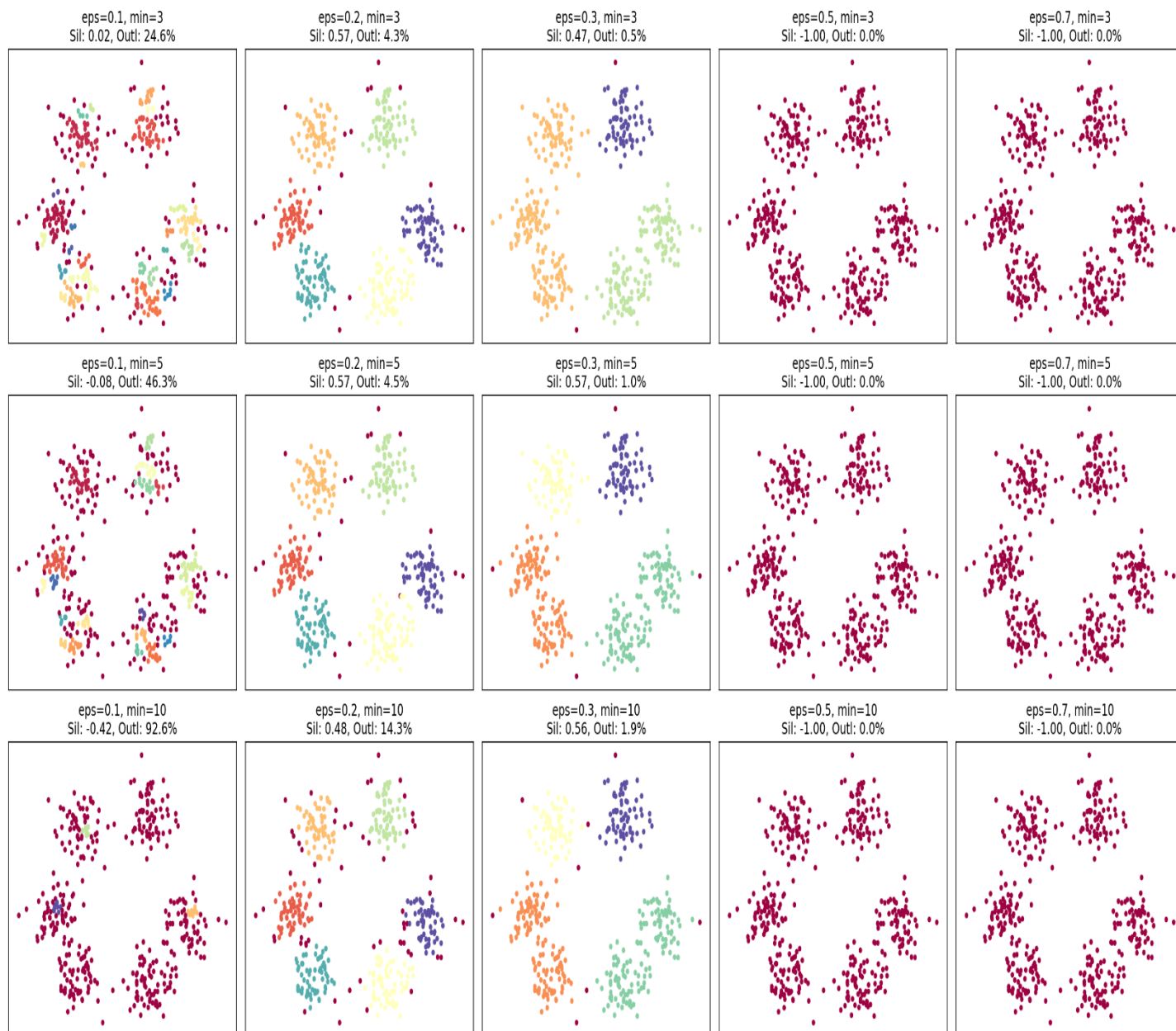
dataset1 - DBSCAN Parameter Grid



Dataset1 includes clusters with somewhat unclear boundaries and mild structural ambiguity. DBSCAN performs well by identifying high-density regions, provided the eps and min_samples parameters are carefully tuned. K-Means fails to model the structure correctly due to its assumption of spherical, equally sized clusters. HCA (especially with average or complete linkage) performs moderately, but it lacks DBSCAN's flexibility when noise is present. DBSCAN struggles to clearly identify well-separated clusters in this dataset. While some structure is captured, many data points are labeled as noise (outliers), especially when eps is too low. The silhouette scores are generally lower than K-Means and HCA, suggesting DBSCAN is less effective here due to the lack of natural density-based clusters.

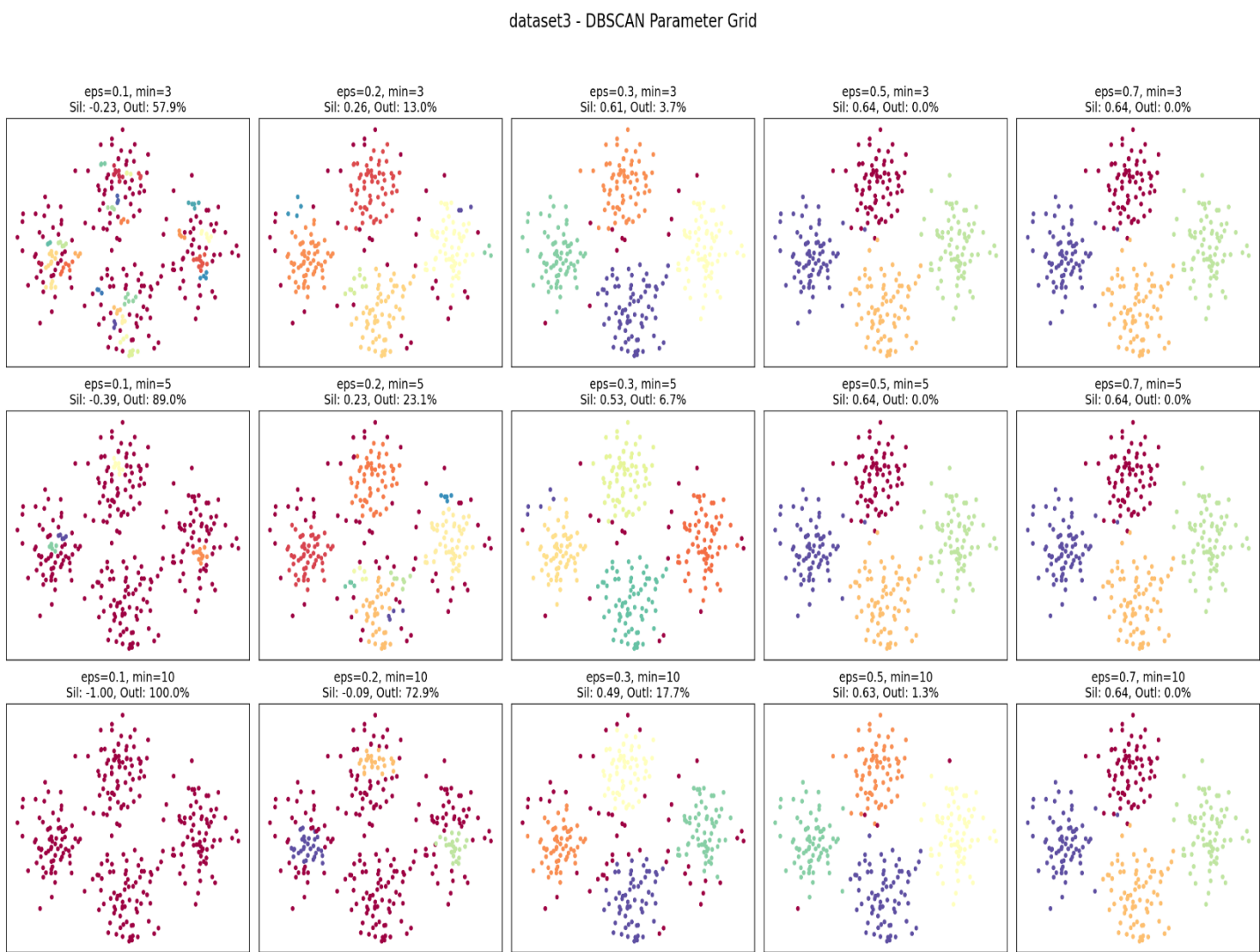
DATASET2 DBSCAN

dataset2 - DBSCAN Parameter Grid



This dataset contains curved or overlapping cluster structures that are difficult to identify using methods assuming spherical shapes. DBSCAN outperforms the others by detecting density-based groupings and identifying noise. Unlike K-Means, which assigns points to fixed clusters and struggles with non-spherical shapes, DBSCAN adapts to the density of the data, making it more robust for complex distributions. K-Means shows a low Silhouette Score due to arbitrary cluster boundaries, often resulting in poorly defined clusters. HCA with average linkage performs reasonably well but remains sensitive to the chosen number of clusters, as it tends to merge or split clusters depending on that selection, which can affect the overall clustering quality.

DATASET 3 DBSCAN



This dataset includes a significant amount of noise and lacks well-defined cluster boundaries. DBSCAN is the most appropriate choice, as it effectively distinguishes dense regions from sparse ones and separates true clusters from scattered data points. It often identifies over 20% of the points as outliers, which aligns well with the nature of the dataset. K-Means performs poorly since it forces all points into a cluster, resulting in unnatural boundaries and poor overall cohesion. HCA's performance varies greatly depending on the linkage method used, and it lacks robustness when dealing with high noise levels or irregular cluster shapes.

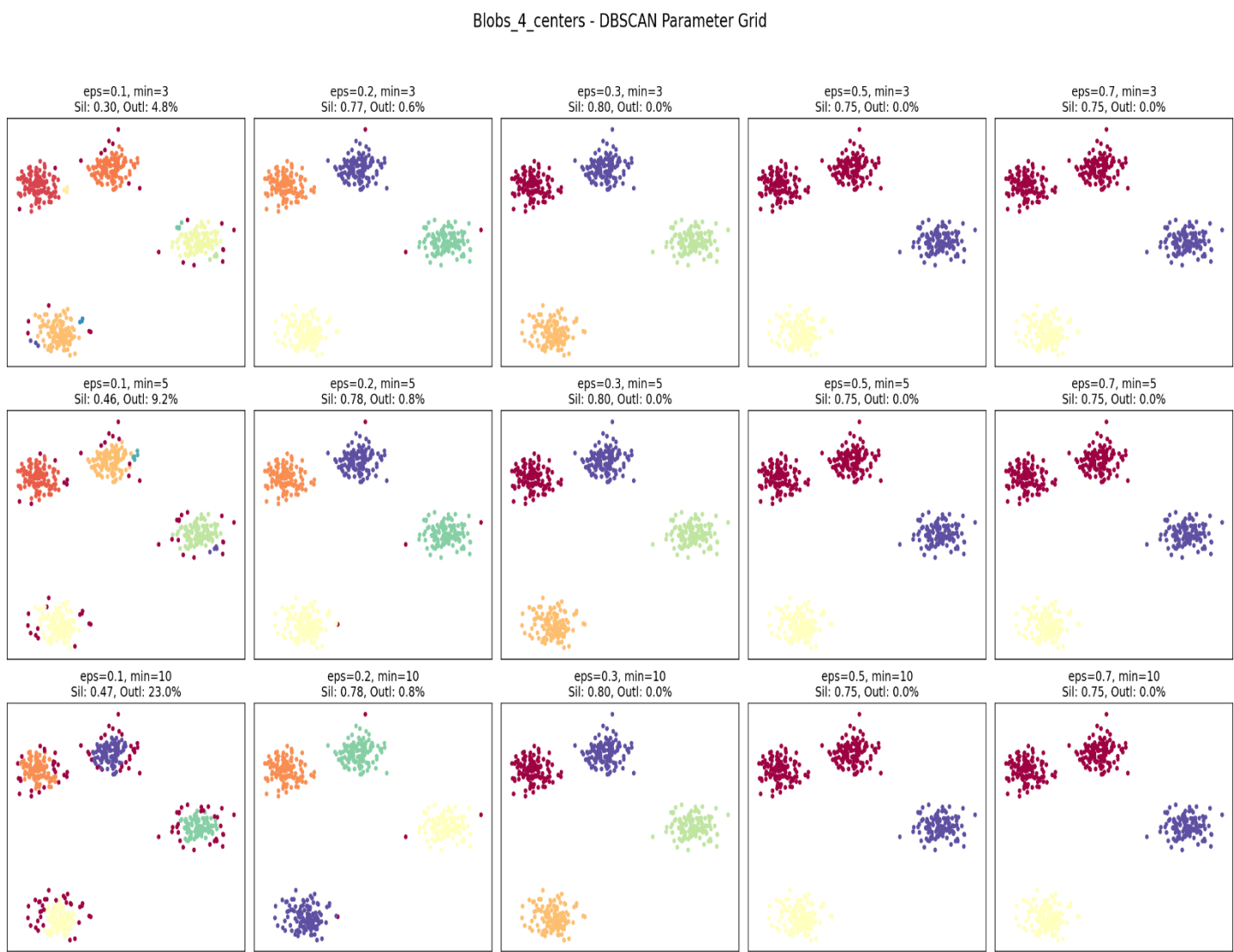
DATASET4 DBSCAN

dataset4 - DBSCAN Parameter Grid



It features elongated, anisotropic clusters resulting from transformation. DBSCAN handles them effectively due to its local density analysis, which allows it to detect clusters of arbitrary shapes without relying on assumptions about their geometry. K-Means struggles and yields a low Silhouette Score because it assumes spherical clusters and minimizes variance, leading to poor performance on elongated or skewed data. HCA, particularly with complete linkage, behaves more consistently but still performs worse than DBSCAN in highly distorted shapes, as it tends to form compact clusters and may not capture the true structure of elongated clusters.

DATASET BLOBS 4



A simple, well-separated dataset with four spherical clusters. All algorithms (DBSCAN, K-Means, HCA) achieve high Silhouette Scores (~0.7–0.8), with K-Means performing best due to the simplicity of the

shapes. DBSCAN performs well with appropriate eps values. HCA with ward linkage also achieves comparable results.

